

External Human Review Packet

Internal status: hostile AI-assisted review completed; external named human sign-off pending.

This packet is designed for a mathematician and a reproducibility reviewer. It does not claim or simulate their approval.

Requested mathematical checks

1. Confirm the one-sided T22 theorem statement and randomized-rule scope.
2. Audit $K_{32}(6) > 0$, the Faà di Bruno/Bell inequalities, Hermite remainder, coefficient enclosures and uniqueness in T16/T30.
3. Check the fixed-line versus arbitrary-node distinction in T27/T37 and the finite-width extension.
4. Check every theorem-scope row against the manuscript.

Requested reproducibility checks

1. Run both Decimal/libmpdec and GMP/MPFR engines from a clean checkout.
2. Confirm 1,421 intervals, 23 chunks, manifest coverage and final ratio.
3. Verify the release ZIP SHA-256 against the externally published digest.
4. Confirm that M146 and M152 are absent from evidentiary claims.

Hostile Review of the Assembled Revision Drafts

Decision: ACCEPT FOR EXTERNAL REVIEW; DO NOT RELEASE AS FULLY VERIFIED UNTIL THE NAMED RELEASE GATES PASS.

Findings

1. Universal-impossibility scope leakage

Pass. The theorem draft is explicitly static and class-scoped. The empirical draft frames failures as tested-family closures and states reopening conditions. The prohibited universal sentences appear only in a “claims removed” section.

2. T22 direction and attainment

Pass. The draft uses a one-sided interval beginning at zero and compares Kerdock with an infimum. It does not claim strict suboptimality or cubature attainment of the auxiliary bound.

3. T16 upgrade

Pass internally, external check required. The draft states the unique all-degree auxiliary optimum and distinguishes it from cubature optimality. The proof outline includes exact quadrature, all-degree reduced costs, sixth-derivative positivity, Hermite feasibility, coefficient positivity, dual equality and uniqueness. The new primal step still lacks a genuinely independent second arithmetic/interval stack.

4. Finite-width wording

Pass after tightening. The finite-width result is stated only for the fixed MUB/Kerdock line universe and under explicit Gaussian-first-layer and even-Hermite nondegeneracy assumptions. Arbitrary-node finite-width T22 remains open. The draft avoids the misleading phrase “finite-width Kerdock optimality.”

5. Signed weights

Pass. T27 is fixed-support; the general signed stability result is labeled weak; off-support signed rules remain open.

6. Nonlinear and adaptive algorithms

Pass. The draft includes a constructive nonlinear ReLU counterexample, a network-dependence warning and residual-kernel recertification. It does not infer all-algorithm Bayes optimality from the limiting kernel.

7. Evidence governance

Pass at manuscript level. M152 is removed, M146 is quarantined, oracle/deployable labels are separated, base-network grouping is required, and a contamination chronology is supplied.

8. Target noise

Pass as limitation, not as dataset-specific quantification. The exact reliability attenuation identity is included. A numerical reliability estimate cannot be supplied from the missing source rows and therefore is not fabricated.

9. Novelty

Pass provisionally. The related-work note identifies the classical ingredients and limits novelty to the kernel-, budget- and estimator-class-specific certificate and support theorems. Priority still requires a human literature review.

10. Reproducibility and AI assistance

Pass as disclosure. The package states that agent agreement is not independent verification and assigns final responsibility to named humans. It does not call the proof formally verified.

Remaining blockers

1. full clean T22 v5.1 regeneration in the final published CI;
2. external publication of the archive digest;
3. independent audit of the T16 sixth-derivative/Hermite/Krawczyk step;
4. named human theorem and reproducibility sign-off;
5. final human related-work and assembled-PDF referee review.

These are external verification requirements, not manuscript wording defects. No further internal prose revision can honestly substitute for them.

Release Gates and Human Review Checklist

The following items are **blocking release gates**, not optional improvements.

Proof release

- ☐ Run the complete T22 v5.1 regeneration from a clean checkout in the pinned CI matrix.
- ☐ Confirm the resulting 59 manifest entries and archive SHA-256 on every supported runner.
- ☐ Publish the archive digest outside the archive in a signed release, immutable commit, DOI/archival record or equivalent.
- ☐ Implement or commission an independent second-stack audit of T16's sixth-derivative positivity, Hermite feasibility and coefficient enclosure. The current C++ audit covers the reduced-cost recurrence only.
- ☐ Have a named human mathematician sign the T22, T16, T27 and finite-width-extension theorem statements and proof outlines.

Empirical release

- ☐ Preserve base-network grouping and frozen selection chronology in every table.

- ☐ Bundle the five omitted Agent 8 source inputs and the omitted Path-2 source archive, or label those packages derived-output-only.
- ☐ Do not reinstate M146 or M152 unless complete primary packages are recovered and independently reproduced.
- ☐ Quantify target reliability/noise for any learning claim that remains in the paper; otherwise retain only comparative, non-universal wording.
- ☐ Audit official score, FLOP and wall-time claims end to end before using “verified” language.

Manuscript review

- ☐ Verify every numerical constant against `UPDATED_NUMERICAL_CONSTANTS.json`; remove the older escape-threshold values.
- ☐ Confirm that T22 always says “infimum” and one-sided gap.
- ☐ Confirm that the finite-width result is T27/fixed-support only.
- ☐ Confirm that every oracle result is labeled **ORACLE DIAGNOSTIC** and visually separated from deployable results.
- ☐ Confirm that no prohibited universal claim appears.
- ☐ Complete a named human related-work/priority review.
- ☐ Run a final hostile review on the assembled PDF, not only on component reports.

Evidence that cannot be self-certified

An AI system cannot satisfy the named-human sign-off, independent-authorship or external-authentication gates by generating another report. These remain open until performed by outside people or infrastructure.